PlantCLEF 2026 — Techniques That Worked

Cumulative progression: **F1 0.28093 (baseline) → 0.50623 (current best, Exp-16C lock 2026-05-05).**

Below are the techniques that improved leaderboard score, in roughly the order they enter the pipeline. Negative experiments are omitted.

All techniques described here are proposed and implemented by **Alper Erten** (aerten3@gatech.edu). Strategies tested by other teammates are excluded; they are not part of the final competition submission either.

Project GitHub Repository: https://github.com/dsgt-arc/plantclef-2026

Original Proposal (Exp-1, abandoned)

- **Cross-Region Transformer with noisy-student distillation on the 212K LUCAS unlabeled set (Exp-1, original proposal)** — The original research plan was an end-to-end *cross-region transformer*: each quadrat would be sliced into region windows, each region encoded independently by a frozen pretrained ViT to produce region embeddings, and a lightweight self-attention transformer would learn inter-tile spatial / co-occurrence structure across regions and pool to a single quadrat-level multi-label prediction. A teacher-student / noisy-student loop would supervise the student on the ~212K unlabeled "pseudo-quadrat" LUCAS images that PlantCLEF provides for optional adaptation, with the teacher being a frozen ViT + simple tile-aggregation head. The expected gain was +0.01 to +0.03 macro-F1 over standard tiling-and-aggregation baselines.
    - **Why it was abandoned.** The mechanism was sound, but the supervision signal was not: the 212K LUCAS unlabeled adaptation set comes from a substantially different sampling distribution and image-statistics regime than the actual 2105-image PlantCLEF test set (different sites, different camera setups, different ecological mix). Pseudo-labels distilled on LUCAS taught the student to model a distribution the leaderboard does not score against, so any in-domain transfer was washed out by the train vs. test domain gap. Two related symptoms made the failure visible: KL/soft-distillation training over all 7806 classes caused mode collapse (the ~7742 near-zero targets per image overwhelmed the ~64 positive signals, F1 → 0); switching to BCE with negative sampling restored a working signal but only to F1≈0.143, well below what direct tiling on the test set was already achieving. The cross-region transformer idea was shelved and the project pivoted to the tiling-and-aggregation lineage that became Exp-2 onward.

Test Image Preprocessing

- **JPEG round-trip preprocessing (Exp-2)** — After Lanczos-resizing each test image, re-encode it through PIL JPEG at quality=85, 4:2:2 chroma subsampling, then decode back to pixels before tiling. The DINO backbones were pretrained on JPEG-compressed natural images, so passing the test images through one round of compression matches the pixel-statistic distribution they expect. Cheap, no parameters to tune, downside is zero. Δ ≈ +0.002 standalone, compounds with downstream stages.

- **Multiscale non-overlapping tiling, scales 4+5+6 = 77 tiles (Exp-7)** — A single 4×4 grid on a 3000×3000 quadrat fragments individual plants and mixes species. Instead, tile each image at three grid sizes (4×4, 5×5, 6×6), giving 77 non-overlapping 518×518 tiles at three effective source resolutions (~747, 600, 500 px). The downstream `max_prob` aggregator can then pick whichever scale best isolates each species. Coarser scales (1×1, 2×2, 3×3) hurt because whole-image crops produce confident wrong labels; 7×7 is diminishing return. Δ ≈ +0.028 over 4×4 alone.

Aggregation & Fusion

- `max_prob` **per-tile aggregation, T = 1.5 (Exp-4)** — For each species class, take the maximum softmax probability across the 77 tiles as the image-level prediction (with temperature T=1.5 sharpening per-tile peaks). Noisy-OR multiplies tile noise; mean/quantile pooling dilutes strong signals. Max preserves the single-best tile evidence per class independently, and is the only aggregator whose F1 is invariant to tile count — the precondition that lets multiscale work. Δ ≈ +0.039.

- **Temporal** `prior_blend` **across location visits (Exp-4)** — Each physical plot in the test set is photographed multiple times. Build a location prior `P_loc(c) = max over visits of P_visit(c)`, then blend with the current image's prediction: `P_final = (1−α)·P_loc + α·P_image`. Applied only to multi-visit quadrats (~76% of the test set). Lets a weak current photo borrow species evidence from earlier visits at the same plot. Δ ≈ +0.022.

- **Per-model α (Exp-11)** — Each branch of the ensemble gets its own α: ViT+kNN α=0.15 (already has external evidence from kNN, so leans on the location prior), ConvNeXt α=0.30 (no kNN arm, trusts its own per-image prediction more, which preserves architecture diversity for the ensemble). Δ ≈ +0.001.

- **Per-source** `sim_w` **aggregator on RNNB + α=0.30 (Exp-14, Stage 4)** — RNNB sites (saltmarsh / coastal dunes in SW France) show year-to-year species turnover: a lush May visit looks nothing like the bare-sand January visit at the same plot, but `max`-pooled prior was making them produce identical predictions. Replace with a per-image prior aggregator weighted by cosine similarity between the current image embedding and each historical sister visit: similar visits contribute fully, dissimilar ones collapse out. Restricted to RNNB only — applying it globally cratered alpine species counts (10.0 → 4.85). Alpine sources keep `max`. Δ +0.00982.

Classifier Model Ensembling (ViT-L-DINOv2 + ConvNext-L-DINOv3)

- **ConvNeXt-Large DINOv3 finetune, 2-phase (Exp-10)** — `convnext_large.dinov3_lvd1689m` (196M params). Phase 1: freeze the backbone, train only the 7806-way classifier head with Adam, lr=0.01 for 100 epochs (cheap, stable head warmup). Phase 2: unfreeze everything and finetune end-to-end from the Phase 1 EMA weights with AdamW, lr=3e-5, layer-wise lr decay 0.8, drop_path=0.3 (higher reg for the larger model), RandAugment + mixup + cutmix, EMA decay 0.9998, 100 epochs on 4×H200. Provides the second ensemble member needed for architecture diversity.

- **ViT-L Phase 2v3 cooldown (Exp-13)** — `vit_large_patch14_reg4_dinov2.lvd142m` finetuned in two stages following the same head-only → full-finetune protocol, then a 10-epoch cooldown at lr=2e-5 on combined train+val+test data on top of Phase 2. Replaces the earlier ViT-B as the production ViT backbone. Standalone +0.011 F1.

- **ConvNeXt + ViT probability-level ensemble, w=0.77 (Exp-11, "Approach A")** — Run the ViT branch with kNN blending first to get the ViT-side per-image probabilities, then linearly mix in ConvNeXt: `P =`

`0.77·P_vit+knn + 0.23·P_cnxt`. Order matters: blending logits *before* kNN breaks the embedding↔logit alignment the FAISS index assumes (it was built on ViT embeddings only) and loses ~0.02. Both branches use the same 4+5+6 scale scheme — letting ConvNeXt use its own preferred 3+4+5+6+7 introduced disagreements ViT couldn't compensate for. Δ ≈ +0.020.

Retrieval Model Ensembling (FAISS)

- **ViT-only FAISS kNN blend (Exp-9)** — Build an exact-search FAISS `IndexFlatIP` over ~860K geo-filtered ViT training-set CLS embeddings (768-d, L2-normalised so inner product = cosine). For each test tile embedding, retrieve the k=20 nearest training neighbours, softmax the similarities with τ=0.07 to get neighbour weights, and aggregate by class label to form a retrieval distribution `p_knn`. Blend per-tile *before* `max_prob` aggregation: `p_tile = β·softmax(logits/T) + (1−β)·p_knn`. The classifier softmax is biased toward common species (long tail under-trained); the retrieval arm is frequency-agnostic and supplies complementary "look-up" evidence for rare species clusters. Δ ≈ +0.017.

- **Differentiated β: β_multi=0.80, β_single=1.00 (Exp-9)** — Inject kNN only on multi-visit quadrats. Single-visit images don't have the temporal prior to absorb retrieval noise; using kNN there cost up to −0.08 F1. Multi-visit quadrats can tolerate β=0.80 (20% kNN weight) because the location prior smooths out noisy retrievals.

Ecological Masking

- **Aquatic-species habitat-impossibility mask (Exp-14, Stage 12)** — Hand-curate 35 fully-aquatic genera (Posidonia, Zostera, Cymodocea, Lemna, Potamogeton, etc.) and 6 aquatic specific epithets (oceanica, submersa, fluitans, natans, …). Mark any of the 7806 species whose Latin binomial matches as habitat-impossible (91 species), and zero those probabilities post-fusion across all sources. Triggered by inspection: Posidonia oceanica (a marine seagrass) was the most-predicted species on dry-soil RNNB sand-dune quadrats. Pure precision improvement, zero parameters, ecologically defensible. Δ +0.00162.

- **Per-source biogeographic mask on alpine sources (Exp-14, Stage 13)** — On CBN-PdlC and CBN-Pla (alpine summit plots only), zero 5 hand-vetted species that survived Stage 12 (no aquatic match) but cannot occur at alpine elevation: tropical C4 carpetgrass (Axonopus compressus), coastal sea-thrift (Armeria maritima), Mediterranean coastal annuals, etc. Other sources are unchanged. Δ +0.00039 LB but proxy-positive on the full cohort; adopted as new best.

Source-Aware Admission & Demotion

- **N-floor admission with confidence-tiered ratio gate, "N3_r0.5" (Exp-16)** — On count-bound sources where the model under-predicts (GUARDEN, LISAH, OPTMix, RNNB), if fewer than 3 species pass the standard threshold=0.100, admit additional candidates up to 3 — but only those whose probability ratio to the already-admitted top-1 is ≥ 0.5 (so we admit confident near-ties, not random low-confidence noise). Doesn't touch sources that already predict enough species (alpine). Largest single-mechanism gain since Stage 4. Δ +0.00557.

- **Habitat-fit demotion using external training-data geography (Exp-16)** — For each (test image, candidate species) pair, compute a habitat-fit factor in [β, 1]. Two components multiplied together: (1) **geo score** — count training observations of that species within 200 km of the source's refined regional centroid; if ≥ n_near_min (40) the species is unpenalised, if 0 it's demoted to β, linear in between; (2)

altitude score — penalise species whose training-data median altitude lies outside the source's expected band. Multiply into the species probability *before* the N-floor admission. Uses external information (PlantCLEF2024 training-data lat/lon/alt) the model never saw at inference. Cosmopolitans (high n_near everywhere) untouched; range-restricted species observed only in distant regions get demoted to β. Δ +0.02361 standalone, final +0.03674 combined with admission retune.

- **Per-source β override on alpine sources (Exp-16C, NEW MECHANISM)** — The single global β was demoting legitimate alpine species too hard, because alpine quadrats were pinned at the topk-10 cap and any extra demotion just dropped real positives below threshold. Override β to 0.45 on CBN-PdlC and CBN-Pla (vs global β=0.15). Tested OPTMix per-source β too — no-op because OPTMix's "out-of-place" species (Plantago subulata-style) have high local n_near and aren't hit by the multiplicative knob. Δ +0.00065 LB, locks final F1=0.50623.

- **Habitat-fit retune at new ViT-L+ConvNeXt-P backbone (Exp-16C)** — When the production backbone changed, the habitat-fit hyperparameters needed re-tuning: n_near_min 30→40, β_global 0.10→0.15. R=200 km, k=10, thr=0.085 unchanged. Final lock 2026-05-05.

Final Submission

- **Fixed global threshold = 0.085 + topK = 10 + SW Europe geo mask** — Hard-threshold per-class probabilities, then cap at 10 species per image. Adaptive per-image thresholds (Exp-6: max-gap, relative-ratio, elbow) all lost to a well-tuned fixed threshold. The geo mask zeroes ~3000 of the 7806 classes that don't occur in southwest Europe. Threshold value drifted across experiments (0.04 → 0.10 → 0.085) only because the probability scale shifts when the pipeline re-blends — the *rule* (fixed global threshold) is unchanged.

PlantCLEF 2026 — README SECTION

End-to-end inference pipeline for the **F1 = 0.50623** PlantCLEF 2026 submission (Exp-16C lock, 2026-05-05).

For a plain-English description of *what* each technique does and *why*, see `Working_Techniques_Summary.md`. This README covers *how to run the code*.

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1. Results

Milestone	Mechanism added	LB F1	Δ
Exp-3 baseline	noisy_or aggregator, T=2.0	0.28093	—
Exp-4	`max_prob` + JPEG q=85 + temporal `prior_blend`	0.35698	+0.07605
Exp-7/8	multiscale tiling 4+5+6 (77 tiles)	0.40100	+0.04402
Exp-9	ViT FAISS kNN blend (β_multi=0.80, β_single=1.00)	0.41754	+0.01654
Exp-11	ConvNeXt-L DINOv3 + ViT probability ensemble (w=0.77)	0.43790	+0.02036
Exp-14	per-source `sim_w` on RNNB + α=0.30	0.45410	+0.01620
Exp-14 Stage 12	aquatic-species habitat mask	0.45572	+0.00162
Exp-14 Stage 13	per-source biogeographic mask on alpine	0.45611	+0.00039
Exp-16 N3_r0.5	N-floor admission + ratio gate	0.46168	+0.00557
Exp-16	habitat-fit demotion (R=200 km, β=0.25)	0.49842	+0.03674
Exp-16C lock	**ViT-L+ConvNeXt cooldown backbones; per-source β override on alpine (β=0.45); habitat-fit retune (n_near_min=40, β_global=0.15)**	**0.50623**	**+0.00781**

Final submission CSV: `${PLANTCLEF_ROOT}/aerten3/submissions/exp16c/exp16c_s12B_alpine_b0.45.csv`

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2. Environment and prerequisites

- **Cluster**: Georgia Tech PACE.
- **SLURM account**: `paceship-clef2026_plantclef`.
- **GPU partitions**: `gpu-rtx6000` (default) or `gpu-h200` (training); QoS `inferno`.
- **CPU partitions**: `cpu-medium`; QoS `embers`.
- **Email for job notifications**: `aerten3@gatech.edu`.
- **Root path** (set before running anything):

```
export PLANTCLEF_ROOT=/storage/home/hcoda1/2/aerten3/ps-clef2026_plantclef-0
```

- **Python deps**: `torch`, `timm`, `faiss-cpu`, `pandas`, `pyarrow`, `numpy`, `Pillow`, `pyyaml`. The PACE conda environment used during the run is pinned in `Exp-10_ConvNext_finetuning/configs/`.
- **Data assumed present**:
  - Test images: `${PLANTCLEF_ROOT}/data/raw/PlanCLEF_test_data/images/`
  - Training images (1.4 M, used for FAISS index + habitat-fit lat/lon): `${PLANTCLEF_ROOT}/data/raw/train_full/`
  - Species map: `${PLANTCLEF_ROOT}/data/raw/class_id_to_species_mapping/class_to_species_id.json`
  - SW-Europe geo mask: `${PLANTCLEF_ROOT}/aerten3/species_masking/sw_europe_mask_bool.npy`

- PlantCLEF2024 training metadata for habitat-fit: `${PLANTCLEF_ROOT}/data/raw/PlantCLEF2024singleplanttrainingdata.csv`

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3. Repository layout

Folder	What it does	Status
`Exp-1_steps_with_lucas_212k_set/`	Original cross-region transformer + noisy-student attempt on the 212K LUCAS set	Abandoned (domain gap)
`Exp-2-JPEG_Compression/`	JPEG round-trip preprocessing	In pipeline
`Exp-4_Enhanced_Tile_Aggregation/`	`MaxProbabilityAggregator`, `prior_blend` temporal fusion	In pipeline
`Exp-7_Multiscale_tiling/`	Multi-scale tile extraction (ViT) at scales 4+5+6	In pipeline
`Exp-9_kNN_Retrieval_Ensemble/`	FAISS index build + per-tile kNN blend (ViT)	In pipeline
`Exp-10_ConvNext_finetuning/`	ConvNeXt-Large DINOv3 2-phase finetune + tile extraction	In pipeline
`Exp-10C_ConvNext_Cooldown/`	10-epoch cooldown of ConvNeXt-L on combined train+val+test	In pipeline (Exp-16C)
`Exp-11_Conv_ViT_Ensemble/`	ConvNeXt + ViT probability-level ensemble ("Approach A")	In pipeline
`Exp-13_ViT_finetuning/`	ViT-L DINOv2 2-phase finetune + Phase 2v3 cooldown	In pipeline (Exp-16C backbone)
`Exp-14_Adaptive_Temporal_Fusion/`	Per-source `sim_w` aggregator + aquatic + biogeographic masks	In pipeline
`Exp-16_Multi_Species_Discovery/`	N-floor admission + habitat-fit demotion	In pipeline
`Exp-16C_HabitatFit_Retune/`	Habitat-fit re-tune at the new ViT-L+ConvNeXt-cooldown backbones; per-source β override	**Final lock**
`Final_pipeline_methods_Summary/`	Detailed methods writeup (frozen at F1=0.4379 — superseded by Exp-16C)	Reference

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4. End-to-end run (final submission)

The minimal command sequence to reproduce **F1 = 0.50623** from raw test images and pre-trained checkpoints. Each line is a SLURM submission; later stages depend on earlier ones.

4.1 Train backbones (skip if checkpoints already exist)

```
# ConvNeXt-Large DINOv3, 2-phase finetune (4×H200, ~3 days total)
sbatch --export=VARIANT=large Exp-10_ConvNext_finetuning/slurm/phase1_train.sbatch
sbatch --export=VARIANT=large,GPU_TYPE=h200,NGPUS=4 Exp-
10_ConvNext_finetuning/slurm/phase2_train.sbatch

# ViT-L DINOv2, 2-phase + Phase 2v3 cooldown
sbatch Exp-13_ViT_finetuning/slurm/phase1_train.sbatch
sbatch Exp-13_ViT_finetuning/slurm/phase2_train.sbatch
sbatch Exp-13_ViT_finetuning/slurm/phase2v3_cooldown.sbatch

# ConvNeXt cooldown (Exp-10C) — same recipe applied to ConvNeXt
sbatch Exp-10C_ConvNext_Cooldown/slurm/cooldown.sbatch
```

4.2 Extract per-tile embeddings + logits at scales 4+5+6

```
# ViT-L tile extraction (3 scales × ~1.5 h on rtx6000)
for SCALE in 4 5 6; do
  sbatch --export=SCALE=$SCALE Exp-13_ViT_finetuning/slurm/extract_phase2v3_rtx6000.sbatch
done

# ConvNeXt-L cooldown tile extraction
sbatch Exp-10C_ConvNext_Cooldown/slurm/extract_4x4.sbatch
sbatch Exp-10C_ConvNext_Cooldown/slurm/extract_5x5_6x6.sbatch
```

Both extractors apply JPEG round-trip (q=85, 4:2:2) before forward pass and write parquet at `${PLANTCLEF_ROOT}/aerten3/parquet/.../scale_{4x4,5x5,6x6}.parquet`.

4.3 Build FAISS kNN index (ViT only, one-time)

```
sbatch Exp-9_kNN_Retrieval_Ensemble/slurm_build_index.sbatch
# Produces: faiss_index_geo.bin, faiss_labels_geo.npy, class_counts_geo.npy
```

4.4 ViT branch: kNN blend + temporal fusion

```
# Per-tile kNN blend (β_multi=0.80, β_single=1.00, k=20, τ=0.07)
# then sim_w aggregator on RNNB + max elsewhere, α_RNNB=0.30, α_default=0.15
sbatch Exp-14_Adaptive_Temporal_Fusion/slurm/run_stage4_simw_BOTH_vitRNNB30.sbatch
```

4.5 ConvNeXt branch: aggregation + temporal fusion

```
# max_prob over 77 tiles, T=1.5, image_topk=64, geo mask
# then sim_w on RNNB + max elsewhere, α=0.30
sbatch Exp-14_Adaptive_Temporal_Fusion/slurm/run_convnext_branch.sbatch
```

4.6 Probability-level ensemble + ecological masks

```
# w = 0.77 ViT+kNN + 0.23 ConvNeXt
# applies aquatic mask (Stage 12) and per-source biogeographic mask (Stage 13)
sbatch Exp-14_Adaptive_Temporal_Fusion/slurm/run_stage13_biogeo.sbatch
```

4.7 Habitat-fit demotion + per-source admission + per-source β (Exp-16C)

```
# Build per-species geographic profile (one-time, ~30 min CPU)
python Exp-16_Multi_Species_Discovery/diagnostics/build_species_geo_profile.py
python Exp-16_Multi_Species_Discovery/diagnostics/refine_source_centroids.py
python Exp-16_Multi_Species_Discovery/diagnostics/add_radius_columns.py

# Final lock: alpine β=0.45 override on top of global β=0.15
sbatch Exp-16C_HabitatFit_Retune/slurm/stage12B_alpine_beta_refine.sbatch
```

The final stage is implemented by `Exp-16_Multi_Species_Discovery/scripts/apply_habitat_fit.py`, invoked as:

```
python Exp-16_Multi_Species_Discovery/scripts/apply_habitat_fit.py \
  --ensemble-parquet  ${PLANTCLEF_ROOT}/aerten3/exp16c/ensemble_phase2v3_cnxt_cooldown.parquet \
  --refined-ref-json
${PLANTCLEF_ROOT}/aerten3/exp16/outputs/habitat_classifier/refined_source_ref.json \
```

```
  --species-geo-csv
${PLANTCLEF_ROOT}/aerten3/exp16/outputs/habitat_classifier/species_geo_profile.csv \
  --geo-mask          ${PLANTCLEF_ROOT}/aerten3/species_masking/sw_europe_mask_bool.npy \
  --aquatic-mask      ${PLANTCLEF_ROOT}/aerten3/exp14/outputs/signals/aquatic_mask.npy \
  --biogeo-json       ${PLANTCLEF_ROOT}/aerten3/exp14/outputs/signals/biogeo_per_source.npz \
  --radius-km 200 --beta 0.15 --n-near-min 40 --use-altitude \
  --per-source-json '{"CBN-PdlC":{"beta":0.45},"CBN-Pla":{"beta":0.45}}' \
  --n-floor 3 --ratio-floor 0.5 \
  --count-bound-sources GUARDEN,LISAH,OPTMix,RNNB \
  --threshold 0.085 --topk 10 \
  --out-csv ${PLANTCLEF_ROOT}/aerten3/submissions/exp16c/exp16c_s12B_alpine_b0.45.csv
```

This is the single command that produces the **F1 = 0.50623** submission CSV.

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5. Final hyperparameter manifest

```yaml
preprocessing:
  jpeg_quality: 85
  jpeg_subsampling: 1   # 4:2:2
  resize: lanczos

tiling:
  scales: [4, 5, 6]
  tiles_per_image: 77
  tile_size: 518

vit_branch:
  backbone: vit_large_patch14_reg4_dinov2.lvd142m  # Exp-13 phase2v3 cooldown checkpoint
  temperature: 1.5
  aggregator: max_prob
  image_topk: 64
  knn:
    index_type: IndexFlatIP
    k: 20
    tau: 0.07
    beta_multi: 0.80
    beta_single: 1.00
    blend_point: per_tile_pre_aggregation
  temporal_fusion:
    aggregator_per_source: {RNNB: sim_w, default: max}
    alpha_per_source: {RNNB: 0.30, default: 0.15}

convnext_branch:
  backbone: convnext_large.dinov3_lvd1689m  # Exp-10C cooldown checkpoint
  temperature: 1.5
  aggregator: max_prob
  image_topk: 64
  temporal_fusion:
    aggregator_per_source: {RNNB: sim_w, default: max}
    alpha: 0.30

ensemble:
  weights: {vit_plus_knn: 0.77, convnext: 0.23}

masks:
  geo: sw_europe_mask_bool.npy
  aquatic: 91 species (genus + epithet pattern match)
  biogeo_alpine: 5 species zeroed on CBN-PdlC, CBN-Pla

habitat_fit:
  beta_global: 0.15
  per_source_beta:
    CBN-PdlC: 0.45
```

```
    CBN-Pla:  0.45
  radius_km: 200
  n_near_min: 40
  use_altitude: true

admission:
  n_floor: 3
  ratio_floor: 0.5
  count_bound_sources: [GUARDEN, LISAH, OPTMix, RNNB]

submission:
  threshold: 0.085
  topk: 10
```

6. Key file index

Artefact	Path
Final submission CSV	`${PLANTCLEF_ROOT}/aerten3/submissions/exp16c/exp16c_s12B_alpine_b0.45.csv`
ViT-L Phase 2v3 cooldown checkpoint	`${PLANTCLEF_ROOT}/aerten3/checkpoints/exp13_vitl/phase2v3_cooldown/model_best.pth.tar`
ConvNeXt-L cooldown checkpoint	`${PLANTCLEF_ROOT}/aerten3/checkpoints/exp10c_convnext_cooldown/model_best.pth.tar`
FAISS index + labels	`${PLANTCLEF_ROOT}/aerten3/exp9_knn/faiss_index/{faiss_index_geo.bin, faiss_labels_geo.npy}`
Geo mask (SW Europe)	`${PLANTCLEF_ROOT}/aerten3/species_masking/sw_europe_mask_bool.npy`
Aquatic mask	`${PLANTCLEF_ROOT}/aerten3/exp14/outputs/signals/aquatic_mask.npy`
Per-source biogeographic mask	`${PLANTCLEF_ROOT}/aerten3/exp14/outputs/signals/biogeo_per_source.npz`
Refined source centroids	`${PLANTCLEF_ROOT}/aerten3/exp16/outputs/habitat_classifier/refined_source_ref.json`
Species geo profile	`${PLANTCLEF_ROOT}/aerten3/exp16/outputs/habitat_classifier/species_geo_profile.csv`
ViT tile parquets	`${PLANTCLEF_ROOT}/aerten3/parquet/plantclef_2026_test/exp13_phase2v3/scale_{4x4,5x5,6x6}.parquet`
ConvNeXt tile parquets	`${PLANTCLEF_ROOT}/aerten3/parquet/exp10c_convnext_cooldown/scale_{4x4,5x5,6x6}.parquet`

7. Reference documents

- `Working_Techniques_Summary.md` — what each working technique does and why.
- `Final_pipeline_methods_Summary/Final_Pipeline_Methods_Summary.md` — detailed methods writeup at the F1=0.4379 milestone (superseded by Exp-16C but still the reference for stages 1–11).
- Per-experiment results: `Exp-*/docs/*Results*.md` — every stage in every experiment is logged with its LB F1, hyperparameters, and the reasoning behind adoption or rejection.

- GitHub repository: https://github.com/dsgt-arc/plantclef-2026